

Pol Roca Cerdá

Email: polrocacerda@gmail.com

Mobile: +34 670974924

Github: github.com/midnaaaaaa

LinkedIn: www.linkedin.com/in/pol-roca-cerdá/

EDUCATION

Universitat Politècnica de Catalunya (UPC - FIB)	Barcelona, Spain
• <i>Master of Science in Innovation and Research in Informatics (MIRI)</i> <i>Specialization in Computer Graphics and Virtual Reality</i>	Sep 2024 – July 2026
Universitat Politècnica de Catalunya (UPC - FIB)	Barcelona, Spain
• <i>Bachelor of Science in Computer Engineering</i> <i>Specialization in Computer Science</i>	Sep 2020 – Jun 2024

EXPERIENCE

Hospital Universitario Valle de Hebrón	Barcelona, Spain
• <i>C++ Software Engineer and Research Intern</i>	Sep 2025 – July 2026
◦ Engineered a medical imaging platform (C++, Qt, VTK, OpenGL) akin to 3D Slicer, integrating real-time 3D volume rendering, synchronized multi-panel views, and interactive volumetric editing tools, used by medical professionals	
◦ Developed a specialized computer vision pipeline for colon segmentation in CT scans , utilizing a custom A* search algorithm for robust medial axis extraction.	
◦ Formulated an advanced A* cost heuristic that integrates anatomical parameters, raw medical image intensities, and tubularity filtering to navigate complex structures and occlusions.	
◦ Implemented the final volumetric segmentation using Graph Cuts optimization, computing custom graph weights derived from the medial axis spatial priors and a 3D watershed map.	
Research Center for Visualization, Virtual Reality and Graphics (ViRVIG)	Barcelona, Spain
• <i>Graphics Programming Intern</i>	Oct 2024 – Sep 2025
◦ Engineered a real-time massive environment viewer in Unity , leveraging compute shaders , GPU instancing , and dynamic LOD for optimal rendering performance	
◦ Implemented advanced rendering techniques , including chunk-based terrain streaming, frustum culling, triplanar mapping, and custom water shaders	
◦ Built a procedural generation pipeline in Python from OpenStreetMap data, modeling terrain path costs via feature histograms (e.g., slope, roughness) for custom A* pathfinding	
◦ Generated dynamic road networks using graph algorithms , evaluating path variations based on difficulty predictions from a trained Random Forest machine learning model	
◦ Adapted the rendering pipeline for Meta Quest VR , optimizing performance and integrating custom post-processing effects like distance-based fog	
◦ Awarded Best Undergraduate Final Project of 2025 at the Spanish Computer Graphics Conference (CEIG 2025 – Eurographics)	
EY (Ernst & Young) - Technology Consulting	Barcelona, Spain
• <i>Fullstack Intern (Part-time)</i>	Nov 2023 – Mar 2024
◦ Developed backend services and frontend components using Java Spring Boot and Vue.js for a large-scale platform	
◦ Maintained unit tests to ensure system reliability and managed CI/CD pipelines with Jenkins	

PROJECTS

• Night Knight – 2D Game & Custom Engine: Developed a 2D game and its custom engine from scratch using modern C++ and OpenGL, implementing sprite animation systems, post-processing framebuffers, and proprietary level-building tools.
• GPU Path Tracer: Implemented a hardware-accelerated path tracer simulating global illumination, in Unity with multi-bounce light transport and importance sampling techniques.
• SpaceVR – Zero Gravity VR Game: Created an immersive physics-based VR experience for Meta Quest using Unity, featuring 6DOF object interactions, dynamic gravity control, and standalone rendering optimizations.
• Physics Simulation Engine: Built a real-time physics engine from scratch (C++, Qt, OpenGL), implementing custom numerical integrators to simulate complex fluid and cloth dynamics.

SKILLS SUMMARY

• Graphics & Engine: Unity, OpenGL, GLSL/HLSL, CUDA, Compute Shaders, GPU Instancing
• Programming Languages: C/C++, C#, Python, Java, JavaScript, SQL, Haskell, Erlang, Prolog
• Libraries & Frameworks: Qt, VTK, ITK, OpenMP, MPI, Scikit-learn, GDAL, NetworkX
• Tools & Architecture: Git, CMake, PlasticSCM, CI/CD, Linux, Windows
• Languages: Native: Spanish, Catalan; Proficient: English